

Wireless Networks

Module 30

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my dear students hello I welcome you all in our series of lectures on cyber security today we are going to study about how to identify basic topologies and their variations we'll also learn about how to determine appropriate topology for a given Network plan so let us begin with it so the first thing is what is a wireless network a wireless network is any type of computer network that uses wireless data connections for connecting network nodes Wireless networking is a method by which homes telecommunication networks and Enterprise installations avoid the costly process of introducing cables into a building or as a connection between various equipment locations Wireless telecommunication networks are generally implemented and administered using radio

communication this implementation takes place at the physical layer of the Osa model Network structure now what are the various types of wireless networks wireless networks are categorized into number one Wireless pan number two wireless LAN number three Wireless man and number four Wireless van let us study them one by one number one Wireless pan pans stands for personal area networks Wireless pans interconnect devices within a relatively small area that is generally within a person's Reach For example both Bluetooth radio and invisible infrared light provides a while span for interconnecting a headset to a laptop now what is a Bluetooth Bluetooth is a wireless technology used to transfer data between different electronic

devices the distance of data transmission is small in comparison to other modes of wireless communication this technology eradicates the use of cords cables adapters and permits the electronic devices to communicate wirelessly among each other a Bluetooth Lan is an ad hoc network which means that the network is formed spontaneously the devices sometimes called gadgets find each other and make a network called EP piconet a Bluetooth lan can even be connected to the internet if one of the gadgets has this capability Bluetooth was originally started as a project by the company it is named for Herald blaten the king of Denmark who United Denmark and Norway blaten translates to Bluetooth in English Today Bluetooth technology is

the implementation of a protocol defined by the IEEE 802.15 standard the Bluetooth protocols let the devices find and connect to each other an act called pairing and securely transfers data Bluetooth defines two types of networks piconet and scatternet a Bluetooth Network called a piconet is a small net a piconet can have up to eight stations one of which is called the primary the rest are called the secondaries all the secondary stations synchronize their clocks and hopping sequence with the primary note that a piconet can have only one primary station the communication between the primary and the secondary can be one to one or one to many the figure shows a piconet although a piconet can have a maximum of seven secondaries and additional eight secondaries can be in the park State a secondary in a park state is

synchronized with the primary but cannot take part in communication until it is moved from the park State because only eight stations can be active in a piconet activating a station from the park state means that an active station must go to the park state next is scatternet piconets can be combined to form what is called a scatter net a secondary station in one piconet can be the primary in another piconet this station can receive messages from the primary in the first piconet as a secondary and acting as a primary deliver them to the secondaries in the second piconet a station can be a member of two piconets now coming back to the Bluetooth let us discuss Bluetooth in detail Bluetooth uses a 2.4 GHz ISM band

divided into 79 channels of 1 MHz each Bluetooth uses the frequency hopping spread Spectrum method in the physical layer to avoid interference from other devices or other networks Bluetooth hops, 1600 times per second which means that each device changes its modulation frequency 1600 times per second to transform bits to a signal Bluetooth uses a sophisticated version of FSK called GFSK that is FSK with Gaussian bandwidth filtering GFSK has a carrier frequency bit one is represented by frequency deviation above the carrier bit zero is represented by a frequency deviation below the carrier the frequencies in megahertz are defined according to the formula $f_c = 2402 + n$ for n is equal to 0 1 2 3 up to

78 for each channel for example the first channel uses carrier frequency 242 MHz and the second Channel uses carrier frequency 243 MHz the link control or Baseband layer is somewhat analogous to the MAC sub layer but also includes elements of the physical layer it deals with how the master controls time slots and how these slots are grouped into frames the link manager handles the establishment of logical channels between devices including power management pairing and encryption and quality of service it lies below the host controller interface line this interface is a convenience for implementation typically the protocols below the line will be implemented on a Bluetooth chip and the protocols above the line will be implemented on the

Bluetooth device that hosts the chip the link protocol above the line is L2 CAP called logical link control adaptation protocol it frames variable length messages and provides reliability if needed many protocols use L2 CAP such as the two utility protocols that are shown the service Discovery protocol is used to locate services within the network the radio frequency

communication protocol emulates the standard serial Port found on the PCS for connecting the keyboard mouse and modem among other devices the top layer is where the applications are located the profiles are represented by vertical boxes because they each Define a slice of the protocol stack for a particular 8 minutes, 1 second purpose specific profiles such as the headset profile usually contain only those protocols needed by that

application and no others for example profiles may include L2 cap if they have Packers to send but skip L2 cap if they have only a steady flow of audio samples now let us discuss about frame format of Bluetooth here in the figure we have various fields such as number one access code this 72 bit field normally contains synchronization bits and the identifier of the primary to distinguish the frame of one ponet from another number two header this 54 bit field is a repeated 8bit pattern it is further divided into six subfields the subfields are number one address the 3-bit address of field can Define up to 7 secondaries if the address is zero it is used for broadcast communication from the primary to all the

secondaries number two type the 4bit type sub field defines the type of data coming from the upper layers number three F this onebit subfield is for flow control when set it indicates that the device is unable to receive more frames that is buffer is full number four a this one bit sub field is for acknowledgement Bluetooth uses stop and weight arq one bit is sufficient for acknowledgement number five s this 1 bit subfield holds a sequence number Bluetooth uses stop and wait arq 1 bit is sufficient for sequencing numbering number six HC the 8bit header error correction of field is a checkm to detect errors in each 18-bit header section the header has three identical 18-bit sections the receiver Compares these

sections bit by bit if each of the corresponding bits is the same the bit is accepted if not the majority opinion rules this is a form of forward error correction this double error control is needed because the nature of the communication via eror is very noisy note that there is no retransmission in this sub layer then comes payload this sub field can be 0 to 2740 bits long it contains data or control information coring from the upper layers now let us see various applications of Bluetooth number one Bluetooth biggest contribution is to provide a phone with a headset that works wirelessly this is possible by providing the collar with an earpiece and a small microphone attached to the caller

shirt the mobile phone can be located in a bag or anywhere in the body the caller can do a number even without touching a button on the mobile phone this technology has the advantage of eliminating the radiation hitting the cerebral region number two PDA PC or laptop which has enabled Bluetooth can communicate with each other and update with its latest information this technology has helped in synchronizing the data easily number three it is difficult to send emails while traveling in a flight on Landing of the flight the Bluetooth enabled laptop can send the email only when it gets in touch with the user's phone number four wireless mouse and keyboards number number five one will be alerted on his or her mobile phone when your laptop receives the

mail number six you can try to locate a printer via laptop you will get the print out of that document once that printer is located so Bluetooth has many applications possible because of the many features that the Bluetooth possess the key features of Bluetooth technology are number one less complication number two less power consumption number three available at cheaper rates and number four robustness now let us discuss next kind of wireless networks called Wireless LAN. Wireless LANs are most important Access Network Technologies in the internet most popular is the IEEE 802.11 Wireless LAN known also as Wi-Fi there are several standards for wireless LAN technology the standard defines two

kinds of services the basic service set BSS and the extended service set ESS now what is basic service set i.e. 802.11 defines the basic service set as the building block of a wireless LAN a basic service set is made of stationary or mobile wireless stations and an optional Central base station known as the access point figure shows two sets in this standard the BSS without an AP is a standalone Network and cannot send data to any other BSS it is called an ad hoc architecture in this architecture stations can form a network without the need of an AP they can locate one another and agree to be part of a BSS a BSS with an AP is sometimes referred to as an infrastructure Network when multiple BSS are connected the stations Within Reach of one another can

14 minutes, 1 second communicate without the use of an AP however communication between two stations in two different BSS usually occurs via two APs the idea is similar to Communication in a cellular network if we consider each BSS to be a BSS and each AP to be a station note that a Mobile station can belong to more than one BSS at the same time next is station types i.e. 802.11 defines three types of stations based on their mobility in a wireless LAN number one no transition number two BSS transition and number three ESS transition Mobility a station with no transition Mobility is either stationary or moving only inside a BSS a station with BSS transition Mobility can move from one BSS to another but the movement is confined inside one

ESS a station with ESS transition Mobility can move from one BSS to another however IEEE 802.11 does not guarantee that communication is continuous during the move then comes MAC IEEE 802.11 defines two MAC of layers the distributed coordination function DCF and point coordination function PCF one of the two protocols defined by it at the MAC sub layer is called the distributed coordination function distributed coordination function uses CSMA/CA as the access method Wireless LAN cannot implement CSMA/CD for three reasons number one for Collision detection a station must be able to send data and receive Collision signals at the same time this can mean mean costly stations and increased bandwidth requirements number two Collision may not be detected because of the Hidden

station problems number three the distance between stations can be great signal fading could prevent a station at one end from hearing a collision at the other end the point coordination function is an optional access method that can be implemented in an infrastructure Network

not in an ad hoc network it is implemented on top of the DCF and is used mostly for time sensitive transmission PCF has a centralized contention free polling access method the AP performs polling for stations that are capable of being pulled the stations are polled one after another sending any data they have to the AP to give priority to PCF over DCF another set of interframe spaces has been defined PIFS and SIFS the SIFS is the same as that in DCF but the PIFS is

shorter than the DFS this means that if at the same time a station wants to use only DCF and an AP wants to use PCF the AP has priority next is frame format the MAC frame consists of 9 fields: number one frame control FC the FC field is two bytes long and defines the type of frame and some control information in all frame types except one this field defines the duration of the transmission that is used to set the value of NAV in one control frame this field defines the ID of the frame number two address there are four address fields each six bytes long the meaning of each 18 minutes, 1 second address field depends on the value of the two DS and from DS subfields number three sequence control

this field defines the sequence number of the frame to be used in flow control number four frame body this field which can be between 0 and 2312 bytes contains information based on the type and the subtype defined in the FC field number five FCS the FCS field is 4 bytes long and contains a CRC32 error detection sequence then comes frame types a wireless LAN defined by IEEE 802.11 has three categories of frames management frames control frames and data frames let us discuss them one by one number one management frames management frames are used for the initial communication between the stations and access points, 1 second points number two control frames control frames are used for accessing the channel and acknowledging

frames number three data frames data frames are used for carrying data and control information next is hidden and exposed station problems let us discuss hidden node problem first as shown in the figure station B has a transmission range shown by the left oval every station in this range can hear any signal transmitted by station B station C has a transmission range shown by the right oval every station located in the range can hear any signal transmitted by C station C is outside the transmission range of B likewise station B is outside the transmission range of C station A however is in the area covered by both B and C it can hear any signal transmitted by B or C assume that station B is sending data

to station A in the middle of this transmission station C also has data to send to station A however station C is out of B's range and transmissions from B cannot reach C therefore C thinks it is free station C sends its data to A which results in a collision at A because this station is receiving data from both B and C in this case we say that station B and C are hidden from each other with respect to A hidden stations can reduce the capacity of the network because of the possibility of the collision the solution to the hidden station problem is the use of the handshake frames figure shows that the RTS message reaches A but not C however because

both B and C are within the range of a the CTS message

which contains the duration of data transmission from B to a reaches C station C knows that some hidden station is using the channel and refrains from transmitting until their duration is over now let us discuss expose station problem now consider a situation that is inverse of the previous one the exposed station problem in this problem a station refrains from using a channel when it is in fact available in the figure station a is transmitting to station B station C has some data to send to station D which can be sent without interfering with the transmission from A to B however station C is exposed to transmission from a it hears what a is sending and thus refrains from sending in other words C is too conservative and was the capacity of the

channel the handshaking messages RTS and CTS cannot help in this case station C hears the RTS from a but does not hear the CTS from B station C after hearing the RTS from a can wait for time so that the CTS from B reaches a it then sends an RTS to D to show that it needs to communicate with d both stations b and a may hear this RTS but station a is in the sending State not the receiving State station B however responds with a CTS the problem is here if station a has started sending its data station C cannot hear the CTS from station D because of the Collision it cannot send its data to D it remains exposed until a finishes sending its data next is Wireless man Wireless metropolitan area network is a type of wireless network that connects several Wireless

lens ymax is a type of wireless man and is described by the IEEE 802.16 standard ymax is basically a technology that enables wireless transmission of data packets at broadband data rate it empowers computer or mobile terminals with enhanced mobility and ability to access high-speed internet without the need of connecting the terminal to any cable network or Wi-Fi hotspot implementation of WMAN requires a similar scale of basic telecommunication infrastructure as built in case of a voice communication Network like GSM and CDMA WMAN base stations sectorized antennas control centers and other critical constituents generally a part of such infrastructure WMAN network has often been claimed in media to be capable of

providing Broadband speeds to a coverage of over 30 Mi with a single base station but that's possible only in ideal conditions practically a single base station can provide a satisfactory Broadband access within a range of 4 to 5 miles with line of sight conditions the coverage can go up to 10 miles rest of the coverage and quality of service details are solely dependent upon the Terrain and population conditions then comes wireless when Wireless wide area networks are wireless networks that typically cover large areas such as between neighboring towns and cities these networks can be used to connect Branch offices of business or as a public internet access 25 minutes, 1 second system the wireless connections between access points are usually point-to-point microwave links using parabolic antennas

on the 2.4 GHz band rather than only directional antennas used with smaller networks a typical system contains base station gateways access points and wireless bridging relays other configurations are mesh systems where each access point access as a relay Wireless technology is also used in cellular telephony and satellite networks cellular telephone is designed to provide Communications between two moving units called mobile stations or between one mobile unit and one stationary unit often called a land unit a service provider must be able to locate and track a caller assign a channel to the call and transfer the channel from base station to the base station as the caller moves out of range to make this tracking possible

each cellular service area is divided into small regions called cells each cell contains an antenna and is controlled by a solar or AC powered Network station called the base station each base station is controlled by a switching office called a mobile switching Center msse the msse coordinates communication between all the base stations and the telephone phone central office it is a computerized center that is responsible for connecting calls recording call information and billing cell size is not fixed and can be increased or decreased depending on the population of the area high density areas require more geographically smaller cells to meet traffic demands that do low density areas once determined cell size is optimized to prevent the interference of

adjacent cell signals the transmission power of each cell is kept low to prevent its signal from interfering with those of other cells last but not the least is satellite Network a satellite network is a combination of nodes some of which are satellites that provide communication from one point on the earth to another a node in the network can be a satellite an earth station or an end user terminal or telephone although a natural satellite such as the moon can be used as a laying node in the network the use of artificial satellites is preferred because we can install electronic equipment on the satellite to regenerate the signal that has lost its energy during travel another restriction on using natural satellites is their distance from the earth which creates a long

delay in communication satellite networks are like cellular networks in that they divide the planet into cells satellites can provide transmission capability to and from any location on Earth no matter how remote this Advantage makes high quality communication available to undeveloped parts of the world without requiring a huge investment in groundbased infrastructure so with this our today's lecture is over today we discussed about the various basic topologies and their variations we also discussed about Wireless standards and their various features apart from their advantages so I hope you understood today's lecture in the next lecture we'll discuss about Security Management till then goodbye thank you